

Syed Fuzail Farhan

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Education

University of Waterloo, Bachelors of Applied Science in Mechatronics Engineering

Technologies

Languages: C++, C, Python, Objective-C, Swift, VHDL, PLC

Hardware/Design: 3D printing, CNC Machining, GD&T, FEA, CFD, Soldering, Engineering Drawings

Software/Tools: Git, Github, Arduino, Django, AutoCAD, SolidWorks, Matlab, STM32, ESP, FPGA, Quartus

Experience

IDEAS Clinic Research Assistant, IDEAS Clinic – Waterloo, ON Aug 2025 – Dec 2025

- Co-designed a 3-day FEA/CFD workshop using ANSYS for 70+ students, achieving a retention rate of over 70%
- Developed industry-backed research case studies (heat exchangers, schlieren imaging, shoulder implant, hands-free crutch) in collaboration with ANSYS to demonstrate real-world simulation applications
- Designed experimental test models in SolidWorks and fabricated them (e.g., 3D printing) to validate simulations, achieving up to 80% agreement between experimental and simulated results
- Assisted a PhD researcher in simulating shoulder replacement surgery using FEA to support and strengthen their research defense
- Designed a 4-day MCU and FPGA workshop for 60+ students, covering Arduino-based micro-controller programming and introductory FPGA design using Quartus Prime
- Developed hands-on lab materials on micro-controller fundamentals (PWM, serial communication, digital I/O), including circuit debugging using multimeter and oscilloscopes
- Began development of an ArcGIS and Python-based dashboard integrating outdoor weather sensors, enabling public data access with admin-level control
- Co-designed a renewable-energy-focused hackathon with faculty and industry partner Brookfield for environmental and civil engineering students

Computation Team Co-op, FormulaNano – Waterloo, ON Jan 2025 – April 2025

- Developed Python pipelines to fully automate UV-Vis and NMR spectra generation, enabling researchers to process batches of samples unattended versus manual step-by-step execution
- Improved usability of in-house quantum chemistry code base by extending the library to handle end to end spectroscopy tasks
- Initiated Raman Spectra Automation by designing the foundational workflow for the vibrational analysis

Software Engineering Intern, GrazeAI – Toronto, ON May 2024 – Aug 2024

- Engineered reusable Python scraper using REST APIs and regex to filter 600,000+ clinical trials by custom criteria (e.g., overnight-stay requirements), reducing manual trial identification from multiple days to minutes
- Created a data filtering system to process large datasets and extract relevant information, contributing to improved data management and analysis
- Automated and generalized data extraction processes, optimizing workflows and reducing manual effort by creating scalable code
- Developed Django full-stack web applications, including a to-do list and polling system, to improve database management efficiency
- Lead a team of 4 to develop a range of generalized data extraction systems
- Contributed to process optimization by improving code efficiency and reducing execution time for various tasks, reducing execution time by 40%